Clements Invitational - Repaired practice edition - Answer key

Division C | Thermodynamics | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

1-13. (1 points) Authored replacement for source item Question 1-13: What is the SI unit of heat energy needed to melt a sample? Distinguish it from the unit of specific latent heat of fusion.

answer: Heat energy is in joules (J). Specific latent heat of fusion is energy per mass, J/kg. The original choices named concepts rather than units. (1 points).

1-20. (6 points) Authored replacement for source item Question 1-20: Solve this case and explain your reasoning: A system receives 500 J heat and does 200 J work. Find delta U and chemistry work.

reasoning: Q=+500 J, Wby=+200 J.
delta U=500-200=300 J.
won=-200 J, giving q+won=300 J. (3 points).

conclusion: Both sign conventions give the same energy change. (3 points).